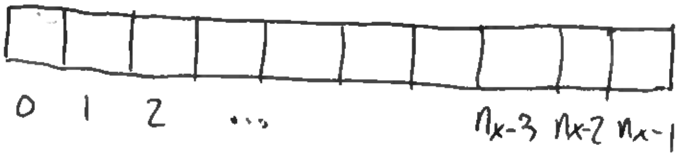
Solutions to exercises in Lecture 7

Q1) Consider the grid



We apply the Neumann boundary conditions at:

* grid point $i=1$: $\dfrac{\partial u}{\partial x} = a$

$$\frac{u_2 - u_0}{2\Delta x} = a \Rightarrow u_0 = u_2 - (2\Delta x)\, a.$$

The Crank-Nicolson algorithm is given by:

$$-r_x \underbrace{u_0^{n+1}} + (2+2r_x)\, u_1^{n+1} - r_x u_2^{n+1} = r_x u_0^n + (2-2r_x) u_1^n + r_x u_2^n$$

substitute with $u_0^{n+1} = u_2^{n+1} - (2\Delta x)\, a.$

so that we have

$$(2+2r_x)\, u_1^{n+1} \boxed{-2r_x u_2^{n+1}} = r_x u_0^n + (2-2r_x)\, u_1^n + r_x u_2^n \boxed{-r_x(2\Delta x)a}$$

↳ change in the matrix

change in the "d" column in our matrix eqn.

* grid point $i = n_x - 2$: $\dfrac{\partial u}{\partial x} = a$

$$\frac{u_{n_x-1} - u_{n_x-3}}{2\Delta x} = a \Rightarrow u_{n_x-1} = u_{n_x-3} + (2\Delta x)a$$

The Crank-Nicolson algorithm is given by:

$$-r_x u_{n_x-3}^{n+1} + (2+2r_x)\, u_{n_x-2}^{n+1} - r_x \underbrace{u_{n_x-1}^{n+1}} = r_x u_{n_x-3}^n + (2-2r_x)\, u_{n_x-2}^n + r_x u_{n_x-1}^n$$

replace with $u_{n_x-1}^{n+1} = u_{n_x-3}^{n+1} + (2\Delta x)\, a.$

so that we have.

$$-2r_x u_{n_x-3}^{n+1} + (2+2r_x)\, u_{n_x-2}^{n+1} = r_x u_{n_x-3}^n + (2-2r_x)\, u_{n_x-2}^n + r_x u_{n_x-1}^n + r_x(2\Delta x)a$$

Q2) For the implicit scheme:

$$\varepsilon_i^{n+1} = \varepsilon_i^n + r_x \left[\varepsilon_{i-1}^{n+1} - 2\varepsilon_i^{n+1} + \varepsilon_{i+1}^{n+1} \right]$$

As in the lecture we can write the error in terms of a Fourier series

$$\varepsilon(x,t) = \sum E_m(t)\, e^{i k_m x}$$

For each k-mode, we have

$$E_m(t+\Delta t)\, e^{i k_m x} = E_m(t)\, e^{i k_m x} + r_x E_m(t+\Delta t)\, e^{i k_m x} \left[e^{i k_m \Delta x} + e^{-i k_m \Delta x} - 2 \right]$$

$$g = \frac{E_m(t+\Delta t)}{E_m(t)} = \left[1 - r_x \left(\underbrace{e^{i k_m \Delta x} + e^{-i k_m \Delta x}}_{2\cos k_m \Delta x} - 2 \right) \right]^{-1}$$

$$g = \left[1 + 2 r_x \left(\underbrace{1 - \cos k_m \Delta x}_{2\sin^2 \frac{k_m \Delta x}{2}} \right) \right]^{-1}$$

$$g = \frac{1}{1 + 4 r_x \sin^2 \frac{k_m \Delta x}{2}}$$

Since the $\sin^2$ term is always positive, $|g| \leq 1$. This means the implicit scheme is unconditionally stable.